

paymio



Services & Solutions





Paymio - Our Services



Products

Paymio adds value to its clients with innovative products developed in the field of payment systems and financial technologies. Paymio's existing product portfolio is designed to meet various financial needs and can be customized according to customer demands.



Consulting Services

Paymio assists its clients in managing their digital transformation processes securely and efficiently with its end-to-end consulting services in financial technologies. From strategy development to implementation, Paymio offers guidance at every stage, delivering solutions aligned with its clients' business objectives and optimizing their financial processes.



Application Hosting and Management Services

Paymio provides comprehensive application management and hosting services to ensure the seamless operation of financial technology solutions. By delivering a high-performance, reliable, and uninterrupted infrastructure, Paymio guarantees that its products are utilized to their full potential.

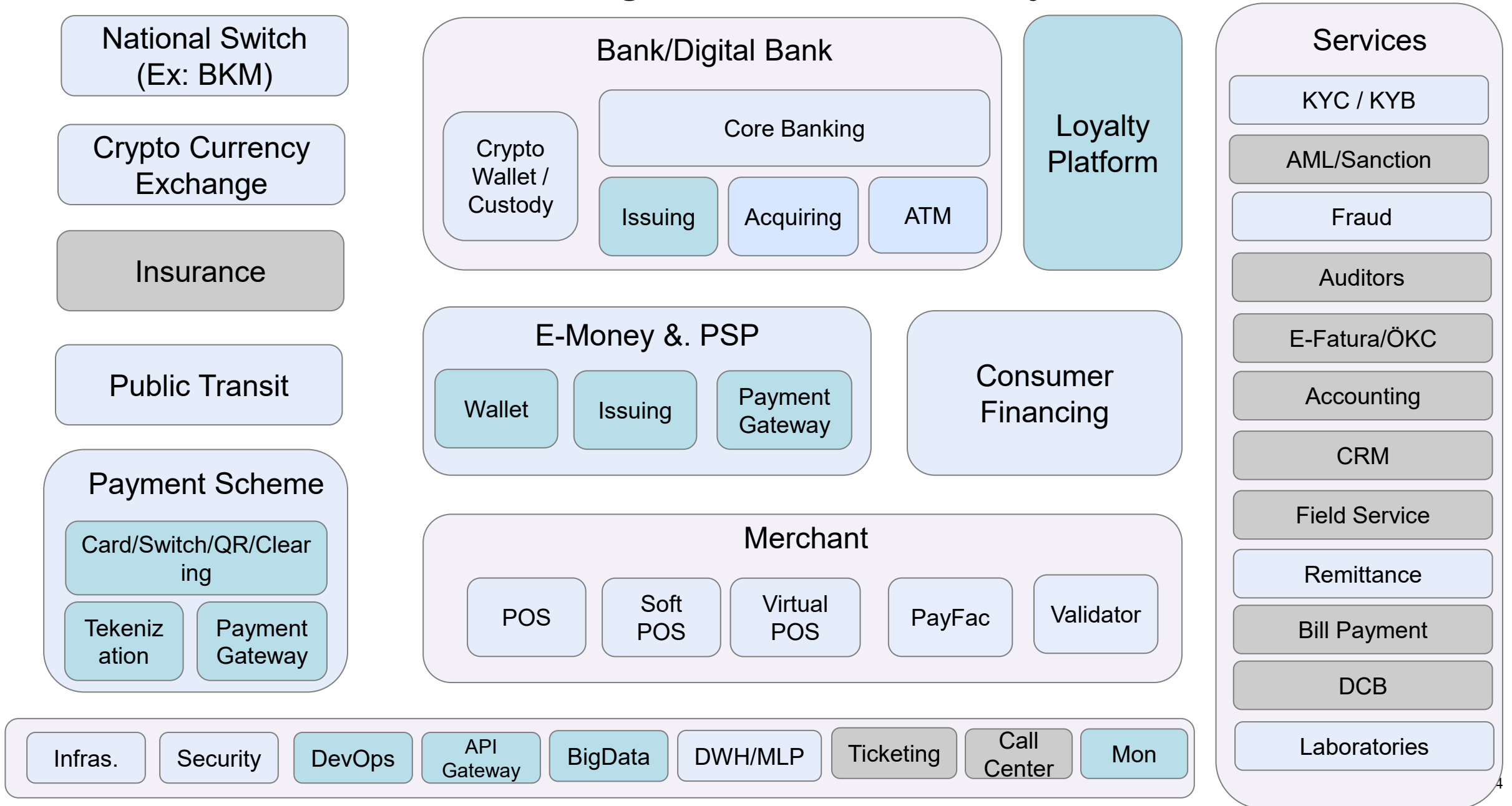


Paymio - Our Fintech Services Roadmap

Target Customers

Product group	Product	     					
		Banks	FinTechs	MNO	Public Institutions	Merchant	Consumer
 National Payment Switch	• Switch • Clearing & Settlement • SYP routing, global network integration						
 Domestic Schema	• Card • QR • Standarts • Integrations						
 Closed Loop Scheme Management	• Transit • Meal Card						
 Card Issuing	• EMV Issuing processing, Personalization • Digital Wallet • Virtual Cards						
 POS & ATM Processing	• POS, vPOS & Electronic Cash Register (ECR) • Acquiring Platform • ATM & Kiosk platform						
 Payment Acceptance	• PSP (Visa/MC/Troy) • Payment Facilitator • National Card • Visa/MC Transit Ready						
 CBDC	• Digital versiyon of Syria SYP • Digital identity and e-KYC						
 Loyalty	• White label Loyalty • Loyalty Platform						

Positioning In Fintech Ecosystem





Current Solutions & Future Developments



Issuer & Acquirer Simulator



- **Load Test: Cards & Terminals**
- **Acquirer System Testing: Visa / Mastercard / Troy/X.. branded cards**
- **Issuer System Testing: Visa / Mastercard / Troy/X.. branded cards**
- Virtual POS Message Simulator (+TechPOS)
- Physical POS Message Simulator
- ATM Message Simulator
- BKM Message Simulator (Issuer & Acquirer)

Product Scope

ISO 8583 Message Builder	Card Definitions	Key Management
EMV Cryptogram Management	Member/Parameter Management	Connection Management
Terminal Management	Simulator Portal	Auto Run & Test Sequences
Scenario Definitions	Customizations for Switch	Reporting



Payment Switch

Payment Switch Capabilities & Architecture



Switch Capabilities



Transaction Routing



Authorization Management



Security & Encryption



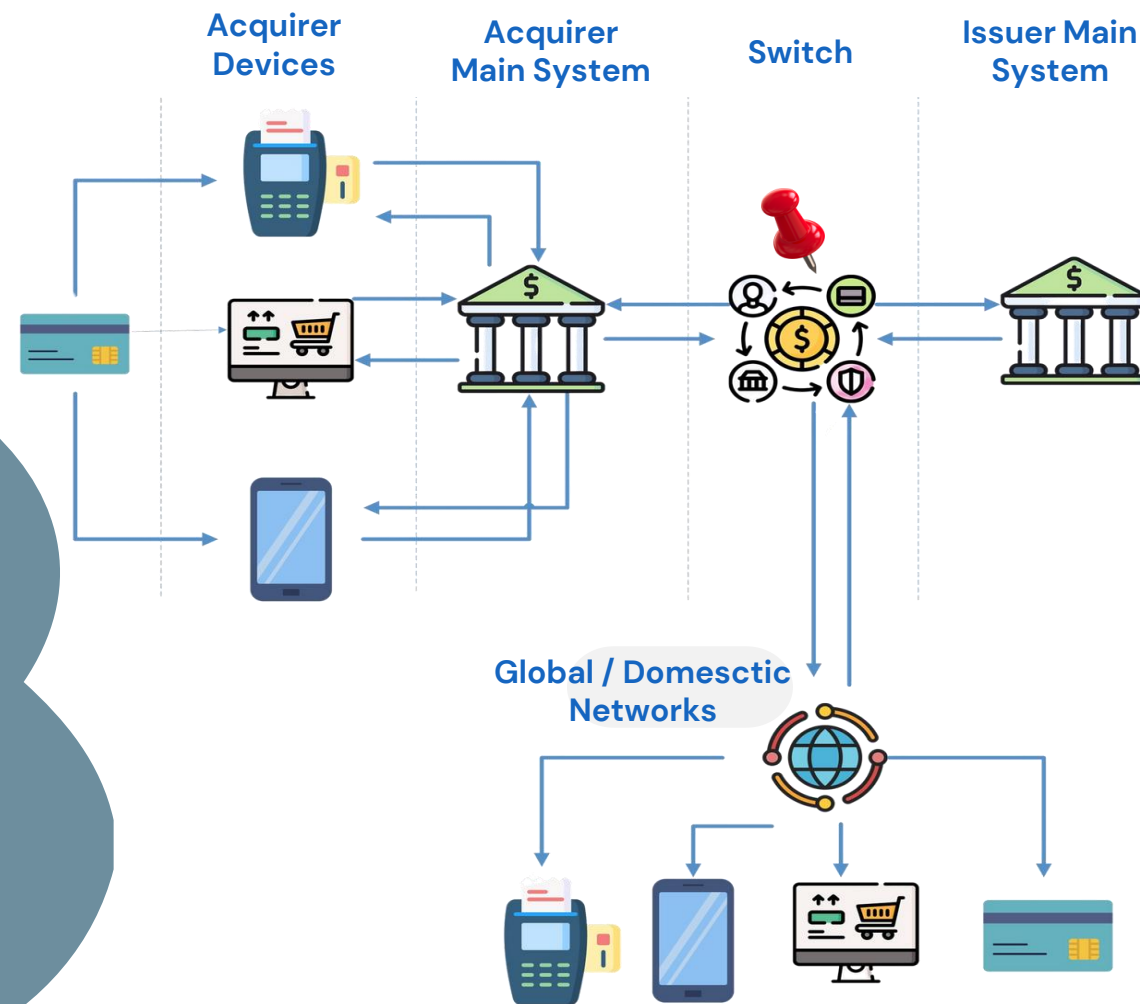
Fraud Detection Integration



Dispute & Reversal Handling

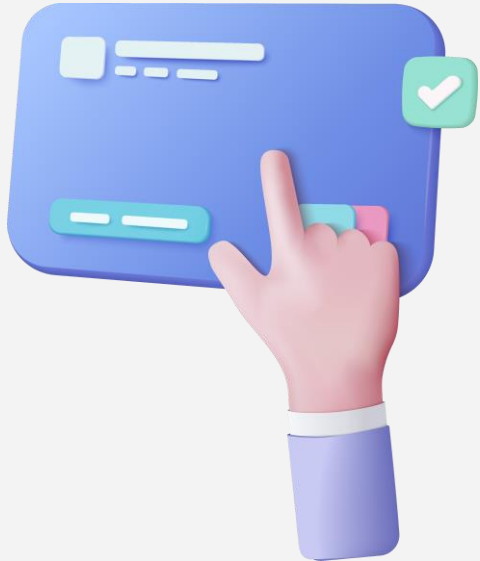
Clearing & Settlement Preparation

Message Transformation





Domestic Payment Schemes



Payment Switch

CA Certification Tool

CA Simulator Test Tool

QR Switch

Tokenization
(TSP & TSM & HCE SDK)

Clearing & Settlement

Domestic Payment Schema
Standard Development

Card Applet / Terminal L2
Kernel





Domestic Scheme

Payment Systems Infrastructure Overview

SPEC Specifications

- **Contact & Contactless Specs**

Technical specifications for both contact and contactless payment methods

- **Personalization Specs**

Card personalization and customization requirements

- **QR Specs**

QR code payment specifications and standards

- **Kernel**

Core payment processing kernel specifications

- **Applet**

Payment application applet requirements

- **HCE**

Host Card Emulation specifications

SW Switch

- **Transaction Switch**

Core transaction switching and routing capabilities

- **Clearing & Settlement**

Financial clearing and settlement processes

- **Fees / Charges**

Fee structure and charging mechanisms

- **Stand-in**

Stand-in processing for offline scenarios

- **ACS**

Access Control Server for 3D Secure authentication

- **QR**

QR code payment processing through switch

CERT Certification & Tools

- **CA Tool and PKI Infrastructure**

Certificate Authority tools and PKI management

- **L2/L3 Test Plan**

Level 2 and Level 3 testing methodologies

- **L2/L3 Tools**

Testing tools for Level 2 and Level 3 compliance

- **Card Test Plan**

Comprehensive card testing procedures

- **Card Test Tool**

Automated card testing and validation tools

- **Perso Validation**

Personalization validation and verification

- **NIV Validation**

Network International Validation processes

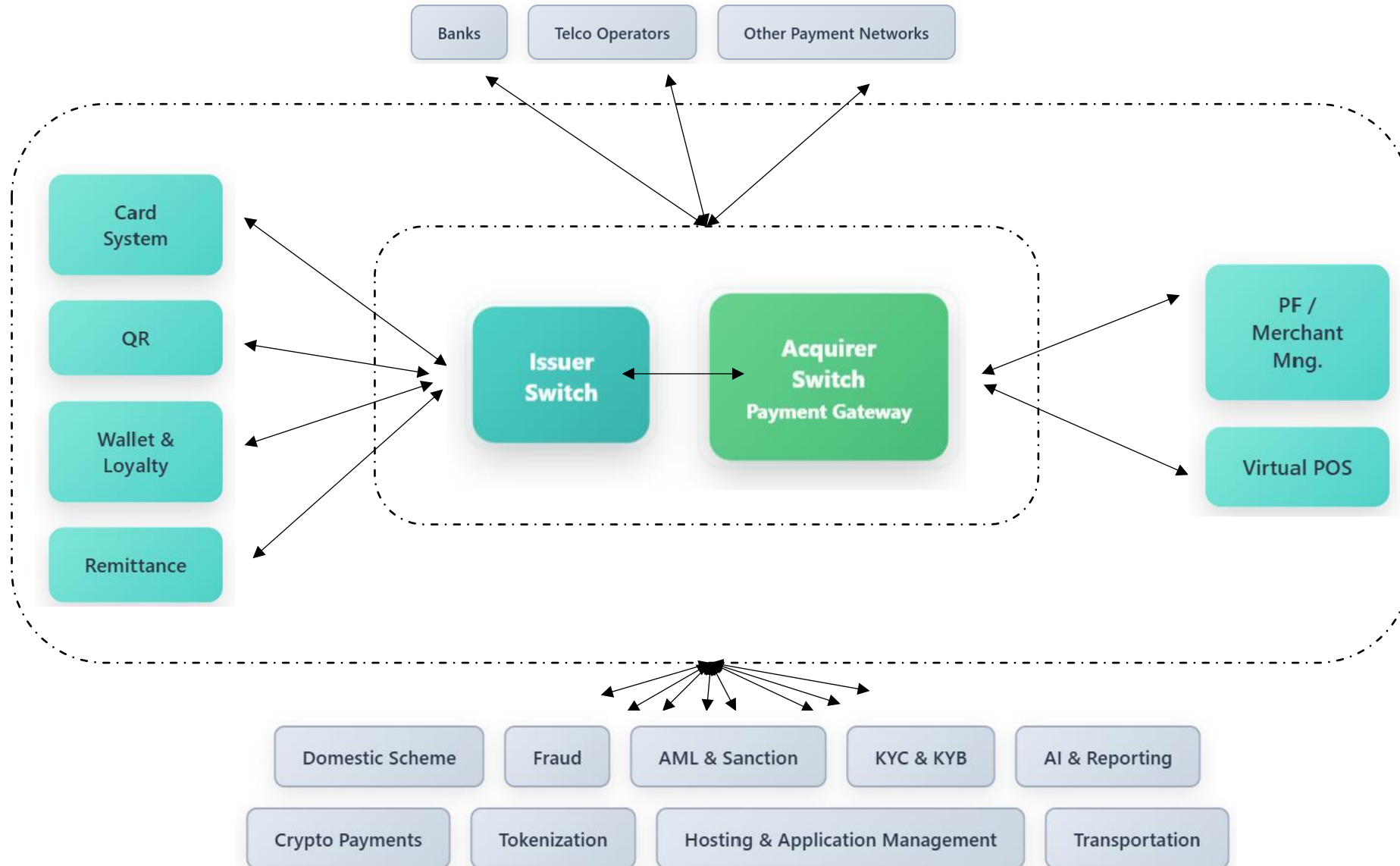
- **Fees/Charges (Certification)**

Certification-related fees and charges structure



Paymio Payment Systems Overview

Comprehensive Payment Ecosystem Architecture





Digital Wallet (Corporate & Consumer)



- One Click Instant Payment
- No Need for Bank Account
- Digital KYC
- Payments (Invoice, TopUp, Purchase etc.)

- QR Transaction Management
- Transaction History
- Secure Payments (3Ds etc.)
- SDK or Whitelabel Applications

Product Scope

Authorization & Processing	Clearing and Settlement	QR Transaction Support
Fraud & Chargeback	Wallet & Bank Integrations	Customer Acquisition
Campaign Management	Card Printing & Personalization Integration	3rd Party Integrations
Certification & Audit & Regulations	Balance, Limit & Commission Management	Co-branded Cards
	Loyalty Integrations	



Acquiring System



- **Payment services for** merchants to accept **Visa / Mastercard / Troy** branded cards
- **Clearing services for** merchants with **Visa / Mastercard / BKM(Turkish National Switch)**
- Merchant And Terminal Management
- Fraud and Chargeback Integration
- Merchant Settlements

Product Scope

**Various Payment Channels
(Physical POS&Virtual
POS&ATM)**

**Clearing and Settlement
Service
Provider(BKM,Visa,Mastercard)**

**Transaction Routing
(Onus,Domestic,International)**

**Different Merchant Commission
Models**

Specialized Merchant Reports

QR Payments

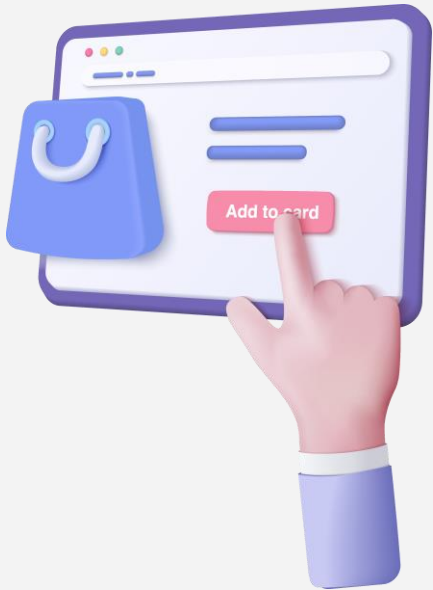
Terminal Management System

3rd Party Integrations

Fraud Integration

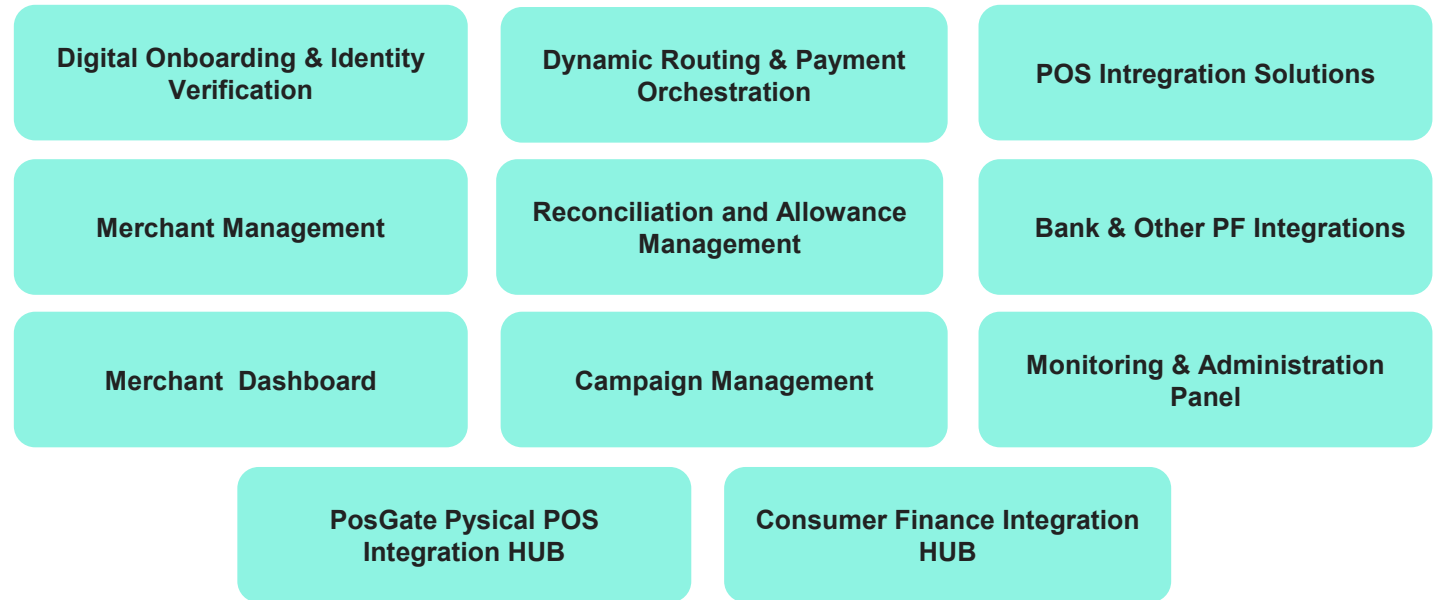


Payment Facilitator



- **End-to-End** payment coverage to both **PSPs and PayFac** with **unique value proposition (VAS and Integrations)**
- Card acceptance via **Android POS, Virtual POS** and **Soft POS**

Product Scope





Card System

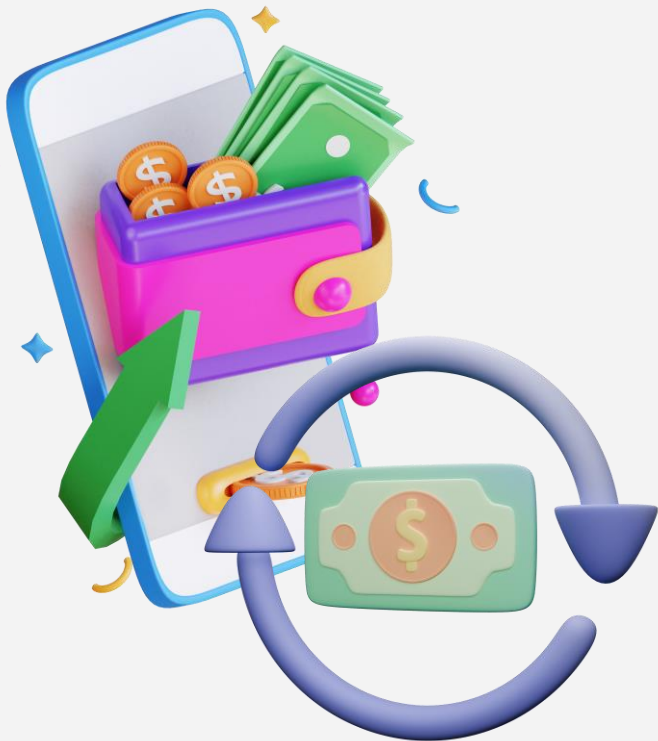
- Prepaid & Debit Card
- Multiple card/payment type support – Magnetic, chip, virtual, contactless and QR Code (EMV Standards), Hybrid Card (Meal & Transportation in a single card), Gift Card, HCE & NFC
- VISA, Mastercard and TROY branded card processing

Product Scope

Authorization & Processing	Clearing and Settlement	QR Transaction Support
Fraud & Chargeback	Wallet & Bank Integrations	Customer Acquisition
Campaign Management	Card Printing & Personalization Integration	3rd Party Integrations
Certification & Audit & Regulations	Balance, Limit & Commission Management	Co-branded Cards



Remittance



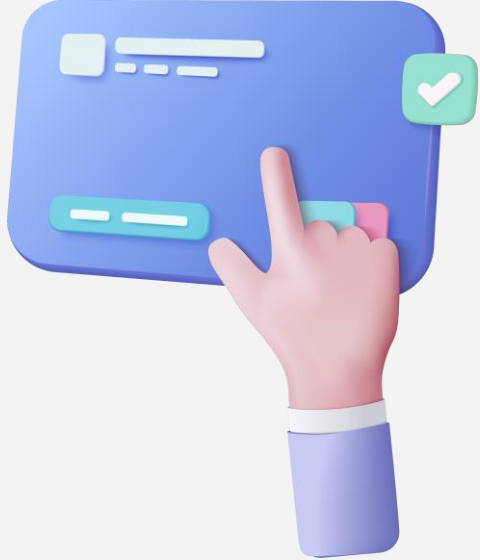
- P2P Money Transfer via Wallet
- Local & Global Remittance Network Integrations (FAST, SWIFT, WIRE etc.)
- Value Added Services (KOLAS, QR etc.)
- Bank Integrations

Product Scope

Real Time Remittance	Clearing and Settlement	QR Transaction Support
Fraud	Wallet & Bank Integrations	Balance, Limit & Commission Management
Campaign Management	Certification & Audit & Regulations	3rd Party Integrations



Card Personalization Systems



Instant Card Issuance

Central Card Issuance

Data Preparation & Key Management





Transportation Systems



Automated Fare Collection
(AFC) System

Acceptance of international
(Visa, Mastercard, Discover)
open-loop cards at public
transportation terminals

Transportation Specific EMV
Applet and Kernel

EMV Card Usage in
Transportation

HCE

QR Payment

Clearing And Settlement
System





Integration Scope

Banks

- Remittance
- Account Management for E-Money
- Merchant Settlement
- VAS Services for Wallet (Investment, Bill Payment, BNPL, Consumer Financing Etc.)

Telco Operators

- Remittance
- Bill Payment
- Direct Carrier Billing

Other Payment Networks

- VISA, Mastercard and Troy Switch for Authorization and Clearing
- MPGS and Cybersource for Merchant Solutions
- BKM for Turkish Domestic Payments
- TCMB for Turkish Domestic Remittance



Shared Products & Services

Domestic Scheme

Fraud

AML & Sanction

KYC & KYB

- **Specifications**
 - Contact & Contactless Specs
 - Personalization Specs
 - QR Specs
 - Kernel
 - Applet
 - HCE
- **Switch**
 - Transaction Switch
 - Clearing & Settlement
 - Fees / Charges
 - Stand-in
 - ACS
 - QR
- **Certification & Tools**
 - CA Tool and PKI Infrastructure
 - L2/L3 Test Plan
 - L2/L3 Tools
 - Card Test Plan
 - Card Test Tool
 - Perso Validation
 - NIV Validation
 - Fees/Charges (Certification)

- Rule Based
- Payment Systems Best Practices
- Real Time Actions
- Near Real Time Actions
- Offline Actions
- Monitoring

- Global & Local Screening
- Transaction Monitoring
- Risk Assessment

- Digital Onboarding
- KPS Integration for KYC
- KPS, KKB, GIB Integrations for KYB



Shared Products & Services

AI & Reporting

- Analytics & Business Intelligence
- Regulatory Reports
- AI Infra for Other Modules

Crypto Payments

- Payment Acceptance
- Crypto Wallet Infrastructure
- P2P Transfer

Tokenization

- TSP
- TSM Backend & Backoffice
- Mobile HCE SDK

Hosting & Application Management

- 7/24 Application & DB Management
- Application & Server & DB Monitoring
- On premises & Cloud Hosting
- Managed Services



System Tool Details



Paymio Payment Systems Tools

Configuration & Cloud Platform



CI/CD & Code Management



Azure DevOps



Microservices Platform



Monitoring & Logging & ITSM





Paymio Payment Systems Tools

Configuration & Cloud Platform

- 1  Azure
- 2  aws
- 3  Google Cloud
- 4  Terraform
- 5  ANSIBLE

- 1 Microsoft's cloud computing platform for building, testing, and managing applications and services.
- 2 Amazon Web Services provides scalable cloud computing services for hosting, storage, and more.
- 3 Google's cloud platform offering infrastructure, data analytics, and machine learning services.
- 4 Infrastructure as Code (IaC) tool for provisioning and managing cloud resources declaratively.
- 5 Open-source automation tool for configuration management, application deployment, and task automation.



Paymio Payment Systems Tools

CI/CD & Code Management

- 1  GitLab
- 2  Azure DevOps
- 3  sonatype
- 4  Azure Registry
- 5  Amazon ECR
- 6  HARBOR
- 7  aqua trivy
- 8  sonarqube
- 9  argo

- 1 DevOps platform offering source code management, CI/CD, and project planning tools.
- 2 Microsoft service providing developer services for planning, collaboration, building, and deployment.
- 3 Manages software supply chains and controls open-source component usage (e.g., Nexus Repository).
- 4 Private Docker container registry for storing and managing container images.
- 5 Amazon's Elastic Container Registry for secure container image management and deployment.
- 6 Open-source container image registry to secure and manage artifacts across cloud-native environments.
- 7 Vulnerability scanner for containers, Kubernetes, and other cloud-native artifacts.
- 8 Tool for continuous inspection of code quality to detect bugs, vulnerabilities, and code smells.
- 9 Kubernetes-native suite of tools for workflows, application deployment, and progressive delivery.



Paymio Payment Systems Tools

Microservices Platform

- 1  docker
- 2  kubernetes
- 3  dapr
- 4  redis
- 5  RabbitMQ
- 6  CAMUNDA
- 7  HashiCorp Vault
- 8  HashiCorp Consul

- 1 Platform for developing, shipping, and running applications inside containers.
- 2 System for automating deployment, scaling, and management of containerized applications.
- 3 Distributed application runtime simplifying the building of resilient microservices.
- 4 In-memory data store used as a database, cache, and message broker.
- 5 Message broker that facilitates communication between distributed systems with queues.
- 6 Workflow and decision automation platform for modeling and executing business processes.
- 7 Tool for securely storing, accessing, and managing secrets and sensitive data.
- 8 Tool for service discovery, configuration, and service mesh by HashiCorp.



Paymio Payment Systems Tools

Monitoring & Logging & ITSM

1



Grafana

2



Stack

3



Prometheus

4



Microsoft Teams

5



1

Platform for visualizing time series data and monitoring system health and metrics.

2

Set of tools (Elasticsearch, Logstash, Kibana) for log analysis and visualization.

3

Open-source monitoring and alerting system designed for reliability and scalability.

4

Collaboration platform combining chat, meetings, and file sharing for teamwork.

5

Messaging app for teams that supports organized communication and integration with other tools.



Referances - Our Team

Financial Institution	Detailed Services	Consulting Services	Payment Services	Software Maintenance Service	Software Operations Service	Hardware Infrastructure/Hosting Service	Security Service	Tarih
BKM	Payment Services Troy Turkia National Payment Scheme ; Specs, Applets, Kernels End-to-End Developments. Mobile Wallet App Tokenization Card Digitization HCE & NFC Payment	✓	✓	✓			✓	2016-2022
Mercury - BAE	Payment Services Mercury National Payment Scheme Mercury National Payment Scheme Domestic Switch System Mercury National Payment Scheme International Switch System All Payment Transactions Simulation Platform Rupay Integration MIR Integration Discover Integration Rwanda Integration Nigeria Integration AEP Integration - Ongoing.. Meeza Integration - Ongoing..	✓	✓	✓	✓	✓	✓	2024 -
NPCI - India	Payment Services Rupay National Payment Scheme ; Specs, Applets, Kernels..	✓	✓	✓			✓	2016-2017
Asis - India	Payment Services Rupay National Payment Transportation Services Kiosk Card Top-Up Transit Card Payments (Validator)	✓	✓	✓			✓	2017-2019
Waricard - Senegal	Payment Services Waricard National Payment Scheme ; Specs, Applets, Kernels..	✓	✓	✓			✓	2016-2016
Card Production - Kenya	Payment Services National Card Production Services, Instant & Mass, Ellams Office	✓	✓	✓			✓	2016-2022
Visa Transit - Urban Mobility	Payment Services Turkia National Transit Switch ; Specs, APIs, Services Transit Switch Systems End-to-End Developments. Transit Company &Municipal & Banks: Sales, Switch, Clearing & Settlement, Accounting, QR, POS, Etc.. Istanbul, İzmir, ... 50+ City & 30K + Terminals All Open Loop Transactions; Visa, Mastercard, Amex, Troy, Etc..	✓	✓	✓	✓	✓	✓	2022-2024



Referances - Our Team

AlbarakaTürk	Payment Services Acquiring Systems End-to-End Developments. All Payment Transactions Simulation Platform 25K + Terminals	✓	✓	✓	✓	✓	✓	2025-
PTT - Türkiye Kart	Payment Services TRCard National Transit Scheme ; Specs, Applets, Kernels.. National Transportation Systems End-to-End Developments. TRCard Top-Up, Sales, Switch, Clearing & Settlement, Accounting, QR, VPOS, Wallet, Etc..	✓	✓	✓	✓	✓	✓	2018-2024
AxisBank-India	Payment Services Mastercard HCE & NFC Payment Mobile Wallet App Tokenization Card Digitization	✓	✓	✓			✓	2017-2018
BEKO	Payment Services ECR & EFT POS Application, RKL & Switch Systems. (1.2 million devices)	✓	✓	✓			✓	2017-2022
TAM ATM	Single ATM for all banks (20K ATM) Ziraatbank, Halkbank, Vakıfbank, PTT, Emlak Bank, Vakıf Katılım Bank...	✓	✓	✓			✓	2022-2024
Common POS	Single POS for all banks (150K POS) Ziraatbank, Halkbank, Vakıfbank Andriod POS BKM Tech POS EFT POS ECR POS	✓	✓	✓				2022-2024
Paycell	E-Money Company Payment Services.. Prepaid Card, VPOS, Andoid POS (50K devices)	✓	✓	✓	✓	✓	✓	2021-2024
PTT	E-Money Company Payment Services.. PTT Issuing & Card Management Platform (7 million cards) PTT Pos & Acquiring System PTT PF Sytem	✓	✓	✓	✓	✓	✓	2019-2024
Belbim	E-Money Company Payment Services.. Transportation Systems End-to-End Developments. Istanbulkart Top-Up, Sales, Switch, Clearing & Settlement, Accounting, QR, VPOS, Wallet, Etc.. (25 Million Cards) Open Loop Transit Systems	✓	✓	✓	✓		✓	2016-2024



Referances - Our Team

Birleşik Ödeme	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓				2016-2017
DGPara	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
İninal	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓		✓	2019-2024
Ozan	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
İyzico	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
Papel	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓				2022-2024
QNB - Kassa	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2020-2024
QNB - Mono	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2020-2024
Aköde	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
TEB	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
Payfix	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
Opet	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓				2023-2024
Petrol Ofisi	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓				2023-2024
TT Ödeme	E-Money Company Payment Services.. E-Money Software, POS, Wallet, PF, Card..	✓	✓	✓	✓	✓	✓	2022-2024
Zubizu	Loyalty Company Payment Services.. Mobile App, BNPL, Wallet, Etc..	✓	✓	✓	✓	✓	✓	2020-2024
Paye	Corporate Meal Card Company Payment Services.. Meal Card Systems End-to-End Developments. Top-Up, Sales, Accounting, QR, VPOS, Wallet, Etc..	✓	✓	✓	✓	✓	✓	2019-2024
Others:Paladyum, MoneyOut, FastPay-Asis, Elekse, MoneyTolia vs..								

Note: These references are from our team. These are financial institutions to which we, as a team, have previously provided consulting, software development, product and service development, infrastructure, and operational services.



Thank You

‘Thank you,